

LAB REPORT

IDS Fundamentals

Introduction to Intrusion Detection Systems and Snort

IDS Fundamentals · Blue Team · Zen Mier

1. Objectives

A firewall controls what traffic is allowed in and out of a network — but it cannot always detect what attackers are doing once they are inside. An Intrusion Detection System (IDS) fills this gap by watching network traffic for patterns that match known attacks and raising an alert when something suspicious is found. This lab introduces what an IDS is, the different types, and how to use Snort — one of the most widely used open-source IDS tools.

- Understand what an IDS do and, importantly, what it cannot do — it detects but does not block.
- Learn the difference between a Host IDS (watches one machine) and a Network IDS (watches all traffic across the network).
- Understand signature-based and anomaly-based detection methods.
- Learn the different operating modes of Snort: sniffer, packet logger, and NIDS mode.
- Write and apply custom Snort detection rules to identify specific types of network traffic.
- Run Snort against a real PCAP file to detect SSH connection attempts and other attack indicators.

2. What Was Done

Understanding IDS Types

The lab covered the two main categories of IDS deployment. A Host IDS (HIDS) is installed on an individual machine and monitors activity on that specific device — useful for detecting changes to system files or suspicious processes. A Network IDS (NIDS) monitors all traffic flowing across the network, making it better suited for catching attacks as they travel between machines. Snort operates as a NIDS.

Detection methods were also covered. Signature-based detection compares traffic against a database of known attack patterns — fast and reliable for known threats but unable to catch zero-day attacks (attacks that have never been seen before). Anomaly-based detection learns what normal traffic looks like and raises alerts when something deviates from that baseline. A Hybrid IDS uses both approaches together.

Snort Modes and Configuration

Snort can operate in three modes depending on the task. In sniffer mode, it simply displays packets on screen — useful for quick inspection. In packet logging mode, it records traffic to disk in PCAP format for later analysis. In NIDS mode, it analyses traffic against a set of rules and generates alerts when matches are found. The main Snort configuration file is stored at `/etc/snort/`, and custom detection rules are written in the `local.rules` file.

```
GNU nano 4.8 /etc/snort/rules/local.rules Modified
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
alert icmp any any -> 127.0.0.1 (msg:"Loopback Ping Detected"; sid:10003; rev:1;)
# This file intentionally does not come with signatures. Put your local
# additions here.
alert icmp any any -> $HOME_NET any (msg:"Ping Detected"; sid:1000001; rev:1;)
alert tcp any any -> $HOME_NET 22 (msg:"SSH Connection Detected"; sid:1000002; rev:1;)
```

Writing and Testing Custom Rules

Custom Snort rules were written and added to the local.rules file to detect specific traffic — including SSH connection attempts and ICMP (ping) activity. Each rule follows a defined structure: action, protocol, source IP and port, direction, destination IP and port, and rule options including a message and a unique rule ID (SID).

Practical Lab — PCAP Analysis

As a forensic investigator scenario, Snort was run against a provided PCAP file (Intro_to_IDS.pcap) containing traffic captured during a real attack. Running Snort with the custom rules produced alerts for two types of activity: an SSH connection attempt from IP 10.11.90.211 to the target machine, and ICMP ping activity. Both were identified through the rule messages triggered in the Snort output.

```
ubuntu@tryhackme:/etc/snort$ sudo snort -q -l /etc/snort -r Intro_to_IDS.pcap -A alert_fast -c /etc/snort/snort.lua
07/18-12:52:59.337559 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:52:59.510662 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:52:59.510663 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:52:59.693619 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:52:59.696547 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:52:59.696584 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:52:59.867507 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:52:59.868435 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:00.050645 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:00.057313 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:00.272189 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:00.444487 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:00.444488 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:00.623967 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:00.647745 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:00.830895 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:00.830918 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:01.364435 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:01.366033 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:01.536660 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:01.536660 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:01.710516 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:01.710517 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:01.710548 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:01.763394 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:08.992981 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:09.167665 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:09.923610 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:10.097583 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:10.292052 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:10.396573 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:10.467526 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:10.571659 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:10.609756 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:10.783691 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:10.783692 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:10.783776 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:10.783976 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:10.783976 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
07/18-12:53:10.784025 ** [1:1000002:1] "SSH Connection Detected" ** [Priority: 0] {TCP} 10.11.90.211:54334 -> 10.10.161.151:22
```

3. Summary

This lab established the foundation for all practical IDS work. Understanding that an IDS only detects and does not block is critical — it generates alerts that an analyst must then review and act on. Snort's rule-based approach means the quality of detection is directly tied to the quality of the rules written, making custom rule creation a core analyst skill.

Area Covered	Key Takeaway
IDS vs Firewall	A firewall controls access; an IDS watches for suspicious behavior and raises alerts without blocking.
HIDS vs NIDS	Host IDS monitors one device; Network IDS monitors all traffic across the network.
Detection Methods	Signature-based catches known threats; anomaly-based catches unusual behavior from a baseline.

Snort Rules

Custom rules written in local.rules — structured by action, protocol, IPs, ports, and options.

PCAP Analysis

Snort detected SSH connection attempt from 10.11.90.211 and ICMP activity in the attack capture.
